BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, MARDAN

SECONDARY SCHOOL (10TH CLASS) EXAMINATION, 2022

PHYSICS

Past Paper 2022 • Paper Code: MRD-10-310522 (A)

Retyped faithfully from the original board paper — same questions, no changes.

Section-A

Time Allowed: 20 Minutes | **Maximum Marks:** 12 | **Paper Code:** MRD-10-310522 (A)
VERSION: B

Note: Note: There are three sections in this paper i.e. A, B & C. Instructions: Please attempt Section-A on the MCQs Answer Sheet only. Only black ball point or marker may be used for shading the relevant circle. Please make sure that there is no cutting, over writing, erasing, or multiple circles shading.

Q1. Choose the correct option i.e. A, B, C, or D.

Q1. If electric field is applied parallel to the direction of electron beam, the electrons will .....

(A) Speed up
(B) Slow down
(C) Deflect
(D) Not change its state

Q2. Which one of the following information storage device is least used these days?

(A) CD/DVD
(B) Hard drive
(C) Flash drive
(D) Floppy

Q3. What type of nuclear decay most often produces the greatest mass and charge loss?

(A) Alpha decay
(B) Beta decay
(C) Gamma decay
(D) Both A and B

Q4. If the pendulum completes exactly 12 cycles in 2 min, what is the frequency of the pendulum?

(A) 0.10 Hz
(B) 0.17 Hz
(C) 6 Hz
(D) 10 Hz

Q5. Slip rings are part of .....

(A) DC motor
(B) AC generator
(C) Transformer
(D) Magnet

Q6. Cathode rays are beam of .....

(A) Protons
(B) Electrons
(C) Ions
(D) Neutron

Q7. The loudness of a sound is most closely related to its .....

(A) Frequency
(B) Period
(C) Wave length
(D) Amplitude

Q8. The human eye forms the image of an object at its .....

(A) Iris
(B) Retina
(C) Pupil
(D) Cornea

Q9. The unit of charge is .....

(A) Farad
(B) Coulomb
(C) Volt
(D) Electron

Q10. The wire made from which of the following material is a conductor of .....

(A) Glass
(B) Rubber
(C) Silver
(D) Silk

Q11. Two resistors of resistances 1 ohm are connected in parallel, the equivalent resistance is .....

(A) 2 Ω
(B) 1.5 Ω
(C) 1 Ω
(D) 0.5 Ω

Q12. A transformer is used for .....

(A) AC voltages
(B) DC voltages
(C) Both DC and AC
(D) Farming

Section-B

Time Allowed: 2:40 Hours | **Maximum Marks:** 32

Q2. Write short answers to any eight (8) of the following parts. All parts carry equal marks.

(i) What are transverse waves? Give three examples.

(ii) Define pitch and quality of sound with examples.

(iii) If the speed of light in kerosene oil is 2.08×10^8 m/s . Calculate the index of refraction.

(iv) Differentiate between potential difference and electromotive force.

(v) Why the pieces of paper initially attracted by charged comb fly away when they touch it?

(vi) What is resistance and in what units we measure resistance?

(vii) State and explain Coulomb's Law.

(viii) Can an electron at rest be set into motion with a magnetic field?

(ix) What are free electrons?

(x) Draw symbol and truth table for NAND and OR gate.

(xi) Which is more likely to expose, a film kept in a cardboard box, α-particles or β-particles? Explain.

Section-C

Maximum Marks: 21

> Note: Answer any three (3) questions. All questions carry equal marks.

Q3.

(a) [4] What is speed of sound? Explain speed of sound in air.

(b) [3] What is the speed of sound in air at −20 °C?

Q4.

(a) [4] Explain total internal reflection of light? What are the conditions for total internal reflection of light?

(b) [3] A telescope has a magnification of 40 and a length of 1230 mm. What are the focal lengths of the objective and eyepiece?

Q5.

(a) [4] What are transformers? Also describe the purpose of transformers in AC circuit.

(b) [3] A transformer steps down a main supply of 220 V AC to operate a 12 V AC lamp. Calculate the turns ratio of the windings.

Q6.

(a) [4] What are background radiations? Describe its major sources.

(b) [3] Write the nuclear equations for the beta decay of ${}_{82}{}^{210}\mathrm{Pb}$ and ${}_{90}{}^{234}\mathrm{Th}$.